BONE AGE ASSESSMENT FROM HAND X-RAY IMAGES USING DEEP NEURAL NETWORKS

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***Özetçe* —Bu çalışmada, el X-ray görüntülerinden otomatik kemik yaşı tahmini için derin öğrenme tabanlı bir yöntem sunulmaktadır. Önerilen yaklaşım, kemik gelişimini temsil eden articular surface ve epiphysis bölgelerini ayrı girdiler olarak ele alan çift dallı bir evrişimsel sinir ağı mimarisine dayanmaktadır. Omurga ağ seçimi sürecinde DenseNet121, ResNet50, EfficientNetB0 ve EfficientNetB3 mimarileri karşılaştırmalı olarak değerlendirilmiş, en iyi performansı veren EfficientNetB0 tercih edilmiştir. Her iki görüntü dalı, ImageNet üzerinde önceden eğitilmiş EfficientNet tabanlı ağlar kullanılarak transfer öğrenme yaklaşımıyla işlenmiştir. Elde edilen görüntü öznitelikleri, bireyin cinsiyet bilgisi ile birleştirilerek kemik yaşı bir regresyon problemi olarak tahmin edilmiştir.**

Model performansı, test kümesi üzerinde Ortalama Mutlak Hata (MAE) metriği ile değerlendirilmiş ve 10.42 ay MAE değeri elde edilmiştir. Sonuçlar, önerilen yöntemin literatürde rapor edilen çalışmalarla uyumlu, etkili ve güvenilir bir çözüm sunduğunu göstermektedir.

Anahtar Kelimeler—kemik yaşı, derin öğrenme, el X-ray, tıbbi görüntüleme, CNN.

***Abstract*—In this study, a deep learning–based method is presented for automatic bone age estimation from hand X-ray images. The proposed approach is based on a dual-branch convolutional neural network architecture that takes the articular surface and epiphysis regions representing bone development as separate inputs. During the backbone selection process, DenseNet121, ResNet50, EfficientNetB0, and EfficientNetB3 architectures are comparatively evaluated, and EfficientNetB0 is selected as the best-performing model. Both image branches are processed using EfficientNet-based networks pretrained on ImageNet within a transfer learning framework. The extracted image features are combined with the individual's gender information, and bone age is estimated as a regression problem.**

Model performance is evaluated on the test dataset using the Mean Absolute Error (MAE) metric, and an MAE of 10.42 months is achieved. The results demonstrate that the proposed method provides an effective and reliable solution consistent with studies reported in the literature.

Keywords—bone age assessment, deep learning, hand X-ray, medical imaging, CNN.

I. Introduction

Bone age assessment plays a critical role in pediatric radiology for monitoring skeletal growth and supporting the diagnosis of growth-related disorders. In clinical practice, bone age is typically estimated through the visual inspection of hand X-ray images using primitive approaches that rely heavily on expert interpretation. These methods are time-consuming and prone to inter-observer variability, which may lead to inconsistent assessments, particularly in clinical environments with high workload.

Recent advances in deep learning, particularly convolutional neural networks (CNNs), have significantly improved performance in medical image analysis tasks. As a result, automated bone age estimation systems based on CNNs have gained increasing attention due to their ability to provide fast, objective, and reproducible predictions. Several studies have demonstrated that focusing on anatomically relevant regions of the hand, rather than the entire image, can further enhance prediction accuracy.

Motivated by these findings, this study proposes a deep learning–based framework for automatic bone age estimation from hand X-ray images. Prior to the selection of the main model, different architectures were evaluated, and their comparative results are presented under the heading "Proposed Method". The proposed approach employs a dual-branch CNN architecture that processes articular surface and epiphysis regions as separate inputs and integrates gender information as an auxiliary feature. Figure 1 presents an example of an articular surface image, while Figure 2 shows an example of an epiphysis image. By combining complementary anatomical information through feature-level fusion, the method aims to achieve robust and reliable bone age predictions. Experimental results show that the proposed model achieves competitive performance compared to existing approaches, demonstrating its potential as a supportive tool for clinical decision-making.

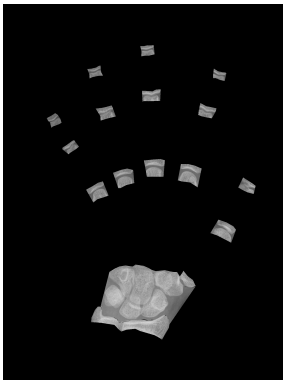

Figure 1 Articular Surface

Figure 2 Epiphysis

II. Proposed Method

Deep learning methods have become increasingly widespread in the field of medical imaging due to the inability of traditional approaches such as the Greulich–Pyle atlas [1] and the Tanner–Whitehouse method [2] to produce reliable results, leading to the development of Convolutional Neural Network (CNN)–based models for bone age estimation. In a study presented by Deng et al., bone age prediction was performed by focusing on the articular surface and epiphysis regions extracted from hand X-ray images [3]. This approach improved accuracy by emphasizing critical anatomical regions of bone development compared to models that utilize the entire hand image. Liang et al. developed a fully automated CNN model that integrates data preprocessing, feature extraction, and regression stages into a single framework, thereby providing a faster and more standardized solution than traditional methods [4]. The BoNet+ model proposed by Guo et al. was specifically designed to generate more stable predictions, particularly for low-quality or degraded images [5].

The proposed framework is designed to estimate bone age from hand X-ray images using a deep learning–based regression approach. Instead of processing the entire hand image, the method focuses on two anatomically significant regions: the articular surface and the epiphysis. These regions are provided as separate inputs to the model in order to capture complementary information related to skeletal development.

Before selecting the final backbone architecture, several widely used deep convolutional neural network models are evaluated, including DenseNet121, ResNet50, EfficientNetB0, and EfficientNetB3. All models are trained and evaluated under identical experimental settings to ensure a fair comparison. The comparative results show that EfficientNetB0 achieves the lowest Mean Absolute Error (MAE) and provides a favorable balance between prediction accuracy and computational efficiency. Therefore, EfficientNetB0 is selected as the backbone network of the proposed architecture.

The network architecture follows a dual-branch convolutional neural network design. Each branch processes one anatomical region independently using an EfficientNetB0 backbone pretrained on ImageNet. Transfer learning is employed to improve convergence and stability, particularly given the limited size of medical imaging datasets. During training, most of the pretrained layers are frozen, and only the final layers are fine-tuned.

Prior to model input, all images are resized to a fixed resolution of 320×320 pixels and normalized according to the requirements of the pretrained network. High-level feature representations extracted from both branches are concatenated and combined with gender information, which is included as an auxiliary input to account for biological differences in skeletal maturation. The fused features are then passed through fully connected layers, and bone age is predicted using a single neuron output layer as a regression value.

Model training is performed using the Mean Absolute Error (MAE) loss function. To prevent overfitting, early stopping and learning rate reduction strategies are applied based on validation performance. This design enables the model to jointly leverage multiple anatomical cues and demographic information for robust bone age estimation.

III. Experimental Results

The performance of the proposed model is evaluated on a held-out test dataset using the Mean Absolute Error (MAE) metric, which is commonly adopted in bone age assessment studies. The dataset is divided into training, validation, and test sets in a subject-independent manner to avoid data leakage.

In order to determine the most suitable backbone architecture, several widely used convolutional neural network models are evaluated and compared, including DenseNet121, ResNet50, EfficientNetB0, and EfficientNetB3. The comparison is conducted using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and the coefficient of determination (R^2) metrics.

Table 1 Model Comparision

Model	MAE(month)	RMSE	R^2
EfficientNetB0	10.42	13.53	0.93
EfficientNetB3	11.55	14.70	0.91
DenseNet121	13.31	18.40	0.86
ResNet50	13.80	17.53	0.87

As shown in Table 1, EfficientNetB0 achieves the lowest MAE and RMSE values while providing the highest R^2 score among the evaluated models. These results indicate that EfficientNetB0 offers superior regression performance and generalization capability for bone age estimation. Therefore, EfficientNetB0 is selected as the backbone network in the proposed framework.

Experimental results show that the proposed method achieves an MAE of 10.42 months on the test set. Considering the inherent variability of skeletal maturation and the regression nature of the task, this result indicates a realistic and competitive level of accuracy. The predicted bone age values follow the general distribution of the ground truth ages, with lower errors observed in mid-range age groups.

Higher prediction errors are observed at very low and very high age ranges, which is consistent with findings reported in previous studies. This behavior can be attributed to data imbalance and increased anatomical variability at the extremes of skeletal development. The use of two anatomically distinct regions and the integration of gender information contribute to improved robustness compared to single-input approaches. The result graph is presented in Figure 3.

Overall, the experimental findings demonstrate that the proposed dual-branch CNN framework provides reliable and generalizable performance for automatic bone age estimation from hand X-ray images.

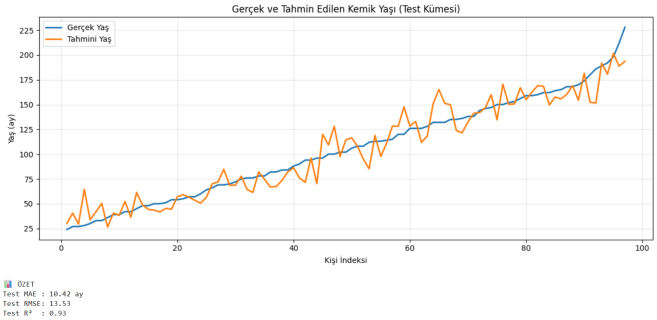

Figure 3 Result Graph

IV. Conclusion

This study uses the dataset in [6], which consists of hand X-ray images. A deep learning–based framework for automatic bone age estimation from hand X-ray images has been presented. The proposed approach employs a dual-branch convolutional neural network architecture that processes articular surface and epiphysis regions separately and integrates gender information as an auxiliary feature. By leveraging transfer learning with EfficientNet-based backbones, the model effectively captures complementary anatomical information relevant to skeletal maturation.

Comparative experiments conducted using different backbone architectures, including DenseNet121, ResNet50, EfficientNetB0, and EfficientNetB3, demonstrate that EfficientNetB0 provides superior regression performance in terms of MAE, RMSE, and R^2 metrics. Based on these quantitative results, EfficientNetB0 is selected as the backbone network of the proposed framework.

Experimental evaluation shows that the proposed method achieves an MAE of 10.42 months on the test dataset, which is consistent with results reported in the literature. While prediction errors increase at the extremes of the age distribution, the overall performance indicates that the model provides reliable and reproducible bone age estimates.

Future work will focus on extending the proposed approach to larger and more balanced datasets in order to improve model generalizability across different age groups and populations. In addition to incorporating additional anatomical regions of the hand, future studies may explore multi-scale and attention-based mechanisms to better capture fine-grained skeletal features. Furthermore, investigating advanced loss functions and uncertainty-aware learning strategies could contribute to reducing prediction errors, particularly at the extremes of the age distribution. The integration of multimodal clinical information, such as growth indicators or hormonal data, may also enhance prediction robustness. Overall, the presented method has the potential to serve as a reliable supportive component in computer-aided clinical decision-making systems and to facilitate more objective and standardized bone age assessment in routine clinical practice.

A subset of records from the results table is presented in Table 2.

Table 2 Test Results Table

ID	Age (months)		
	Actual Age	Estimated Age	Difference
1433	24	30	6
1944	27	41	14
3040	27	30	3
3766	28	65	37
1407	30	34	4
1784	33	42	9
14235	33	50	17
1475	36	26	10
7484	39	41	2
8435	39	38	1
13954	42	52	10
14018	42	36	6
4332	45	61	16
12517	48	49	1
12936	48	44	4
3591	50	44	6
3616	50	42	8
9194	51	45	6
13518	54	45	9
13528	54	57	3
7625	55	59	4
4818	57	57	0
6377	57	54	3
15299	60	51	9
11099	64	56	8
4803	66	70	4
10793	69	72	3
15386	69	85	16
14724	70	69	1
15578	72	69	3
12133	75	78	3
8579	76	64	12
9072	76	61	15
7397	78	82	4
7416	78	75	3
7022	82	67	15
7035	82	67	15
14224	84	74	10
14289	84	82	2
14543	88	86	2
14123	90	76	14
13869	94	72	22
13878	94	96	2
12046	96	70	26
12213	96	120	24
3736	100	109	9
4078	100	128	28
13787	102	97	5

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